



Speak with Your Data

Hackathon Challenge Brief



Background

Data is everywhere — inside CSVs, databases, dashboards, and spreadsheets. Yet most people still need advanced technical skills to extract meaningful insights from it. What if you could simply ask your data a question, just like chatting with a colleague?

With the rise of Model Context Protocol (MCP) servers and large language models, it is now possible to build natural-language interfaces directly on top of structured data — turning plain-English queries into charts, graphs, and visual insights in real time.



The Challenge

Build a conversational data analysis tool powered by an MCP server. Users should be able to upload or connect to a dataset, ask natural-language questions about it, and receive relevant, dynamically generated visualizations as answers — no SQL or coding required.

Objectives

The ideal solution should:

- Allow users to upload or connect to a dataset (CSV, JSON, database, or API).
- Accept free-text questions in natural language, including the ability to specify the type and style of visualization desired (e.g., "Show me sales by region for Q3 as a bar chart with blue colors" or "Give me a donut chart of product categories, sorted by revenue")
- Use an MCP server to interpret the question and generate the appropriate query or logic.
- Automatically produce a relevant visualization (bar chart, line graph, pie chart, heatmap, etc.) based on the answer.
- Handle follow-up or clarifying questions within the same conversation context.
- (Bonus) Support multiple data sources or switching between datasets mid-conversation.

Data & Tools

Data Sources

- Participants will use datasets provided to them, from the repository of the challenge owner

Tools & Platforms

- **Required:** MCP Server
- **Required:** LLM / AI API (e.g., Claude API, OpenAI API, or open-source model)
- **Required:** Visualization library — e.g., Chart.js, D3.js, Plotly, Matplotlib, Recharts
- **Suggested:** Python (pandas, FastAPI) or Node.js for backend logic
- **Suggested:** Streamlit, React, or any frontend framework for the UI
- **Optional:** Jupyter Notebook for exploratory / prototype-level submissions
- **Optional:** Cloud deployment (AWS, GCP, Vercel, Railway, etc.)

Scope & Timeframe

This challenge is designed to be completed within approximately 40 hours of active work. To help teams plan:

Level	What it looks like
Quick & Simple	Jupyter Notebook + local MCP server + static charts from a CSV. Focused on the Q&A flow.
Intermediate	Web app with a chat interface, MCP integration, and dynamic charts rendered in the browser.
Ambitious	Multi-source support, conversation memory, follow-up questions, deployed and publicly accessible.

Good luck — let your data do the talking! 🚀